

Effect of Surface Geometry on Airfoil Performance

Daniel Grover¹  Omar Cochinwala¹  Hamzah Shir¹ 
Ezenbaatar Batjargal¹ 

¹ University of California Berkeley, Mechanical Engineering Department, Berkeley, USA

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Abstract

The surface properties of airfoils have a significant effect on their performance. Understanding the relationship between surface geometry and airfoil performance is crucial in maximizing efficiency of aircraft, an important point in reducing emissions from air travel. This experiment tests the performance of 7 surface geometries on a NACA 0012 airfoil relative to a non-modified airfoil: 3 airfoils with circular dimples, 3 with rectangular grooves parallel to the airfoil length, and 1 with rectangular grooves perpendicular to the airfoil length. After experimenting, it was concluded that adding dimples and cutouts to airfoils significantly reduces the lift they produce, though in some cases drag was also reduced, though by a much smaller amount. It was concluded that a smooth airfoil surface is optimal for performance.

Key words: lift – drag – airfoil – wind tunnel – additive manufacturing

1 Introduction

In the late 19th century, a large change to golf balls was implemented that would double their traveling time when hit. This change involved covering the surface of the golf ball with small dimples. These dimples had significant effects on the aerodynamics and flight trajectories of golf balls. The dimples create a thin turbulent boundary layer, which allows the air to "stick" to the ball longer before separating and flowing to the back, thus decreasing drag on the surface. This phenomena also increases lift because the airflow stays attached longer on the top, and the fast-moving, low-pressure region extends further. This pivot in design helped improve the flight paths of golf balls after being hit, changing how the game was played.

While there is a set standard for the size and number of dimples in professional settings, researchers are constantly studying the effect of changing the number and diameter of the dimples on the aerodynamics and trajectory of a golf ball.

This paper examines at the idea of extending this concept to aircraft. If this drag decrease and lift increase seen on golf balls could be extended to airfoils, aircraft efficiency could be greatly improved. This experiment explores the effects of dimples and grooves on airfoil performance. The hypothesis for this experiment is that adding grooves and dimples will increase an airfoil's performance, namely, increasing its lift and decreasing its drag.

2 Method

The materials used in this experiment were:

- NACA-012 15 mm airfoils with varying surface geometries
- Hesse Hall Draw-Through Wind Tunnel
- Weight set and hanger
- Airfoil mount and strain gauge
- RPM sensor
- Pitot tube
- Pressure sensor
- LabView software

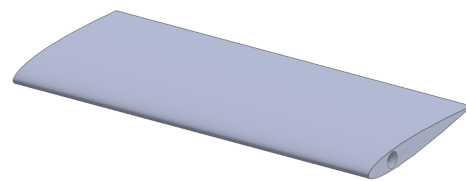


Figure 1. Base Airfoil Design (NACA 0012 15 mm)

The airfoil type used in this experiment was the NACA 0012 - 15mm design, a basic airfoil shape characterized by its symmetric profile. The test articles were additively manufactured using a Creality Ender 3-V3, a fused-deposition modeling (FDM) 3D printer, along with a Bambu A1, another FDM printer. Some test articles were provided by the Hesse Hall Staff, but were also additively manufactured.

The airfoil types tested against the base airfoil are defined as follows (photos can be found in the Appendix):

- Small Dimples - 1081 circular dimples filling the airfoil surface
- Medium Dimples - 253 circular dimples filling the airfoil surface
- Large Dimples - 105 circular dimples filling the airfoil surface
- Small Grooves - 0.125 in grooves in airfoil parallel to airfoil length
- Medium Grooves - 0.25 in grooves in airfoil parallel to airfoil length
- Large Grooves - 0.5 in grooves in airfoil parallel to airfoil length
- Sideways Grooves - 0.5 in grooves in airfoil perpendicular to airfoil length

2.1 Calibration

Before the tests were run, calibration data needed to be collected in order to characterize the response of the instruments used in our experiments. The main data collected



Figure 2. Placing weights on the strain gauge to characterize its response

was from a gauge that measured strain produced by force on the mount where the airfoils were secured in the wind tunnel. To calibrate the gauge, a hanger assembly provided by the Hesse staff was hung at the end of the mount. Ten data points of the voltage output of the strain gauge were then recorded. Weights were then added to the hanger and the voltage at each new weight was recorded.

The strain gauge output was recorded at 1 gram increments from 1 to 10 grams, and then at 10 gram increments from 10 to 100 grams. An additional test was done to avoid the effect of hysteresis on the sensor, where weight was strictly increased from 1 to 100 grams, and then strictly decreased back down to 1. This additional data ensured the calibration curves properly represented the sensor output. During our calibration tests, the strain gauge was oriented at 5 degrees to the horizontal, as measured by a protractor built into the mount. This was the baseline orientation for measurement.

The next sensors that needed to be characterized were those that measured the rotations per minute of the fan powering the wind tunnel and the pressure sensor that measured the speed of wind moving through the tunnel. Using the controller, the fan was set to all possible speeds, and twenty data points were taken at each speed. After collecting this data, it was decided that a speed at about half of the wind tunnel's capability was ideal for our experiments.

This speed was roughly 8.5 meters per second, as determined using Bernoulli's principle and the pressure reading from the wind tunnel's pitot tube. First, the reading was converted from mmHg to Pascals, and then plugged into Bernoulli's relation to determine the wind speed. A room temperature of 68 degrees Fahrenheit and an ambient pressure of about 101,000 Pa was used. This calculation was done with the ideal gas assumption, as air can typically be approximated in this manner and because the exact wind speed was not critical to this experiment, as it was held constant in all tests.

$$V = \sqrt{\frac{2\Delta p}{\rho}} = \sqrt{\frac{2\Delta p}{\frac{p}{RT}}} = 14.85\sqrt{p} \quad (1)$$

This formula was then used to convert the pressure reading to wind speed in meters per second.

Using this calibration data, relationships were determined

Fan Setting vs Wind Speed

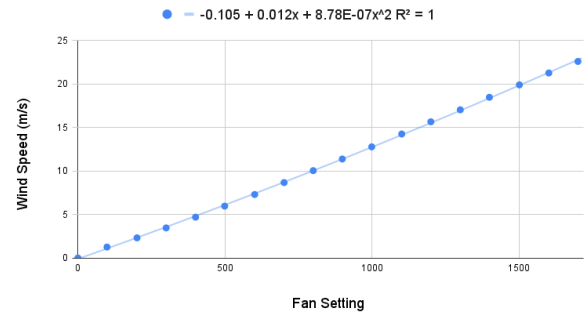


Figure 3. Wind Speed Calibration Data

Strain Gauge Calibration

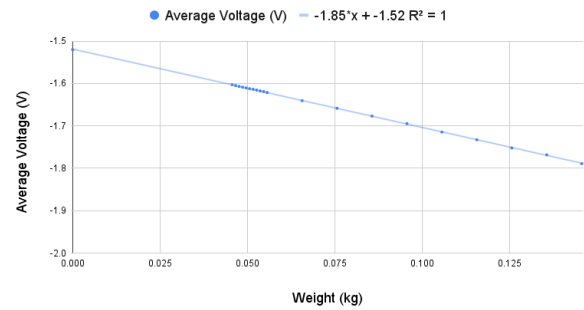


Figure 4. Strain Gauge Calibration Data

between the sensor output and the physical quantities being measured - namely, the force applied to the mount by the airfoils.

- The relationship between weight and voltage was determined to be $V = -1.85w - 1.52$, where w is the weight in kilograms and V is the sensor voltage output.
- The relationship between wind speed and the fan setting was determined to be $S = -0.105 + 0.012f + 8.78 \times 10^{-7}f^2$, where f is the fan setting and S is the wind speed in meters per second.

Drag and Lift Comparison

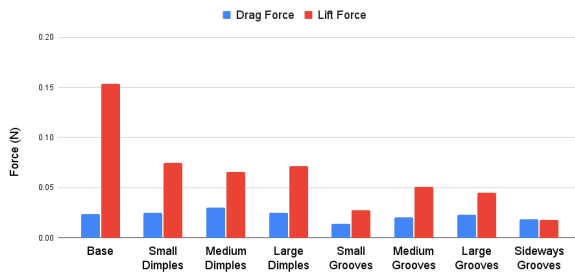


Figure 5. Lift and Drag Comparison

Airfoil Type	Lift Force (N)	Drag Force (N)
Base	0.153	0.023
Small Dimples	0.075	0.025
Medium Dimples	0.065	0.030
Large Dimples	0.072	0.025
Small Grooves	0.028	0.014
Medium Grooves	0.051	0.020
Large Grooves	0.045	0.023
Sideways Grooves	0.018	0.019

Table 1. Lift and Drag data

2.2 Testing

Once the calibration data had been measured, each airfoil configuration was tested. Each airfoil was tested in the same manner to ensure the only changing variable was the surface geometry.

- The airfoil was fastened into the tunnel using the mount provided by Hesse Staff.
- The strain gauge was set to the baseline of 5 degrees.
- The airfoil was set to an angle of 15 degrees to the horizontal as measured by an angle finder.
- The wind tunnel was closed and turned on at our experiment speed of roughly 8.5 m/s.
- 20 data points were recorded using the LabView software after the wind reached our experiment speed.
- The fan was turned off and the strain gauge was rotated 90 degrees clockwise relative to the airfoil to measure drag.
- The fan was turned back on and 20 more data points were taken.

This process was repeated for each airfoil and compared with our calibration data to draw conclusions about the surface geometry's effect on performance.

3 Results

The results of the experiments did not align with the hypothesis. Every configuration tested performed significantly worse than the base airfoil. Every airfoil had significantly reduced lift, and some showed reductions in drag, though the reductions were quite small.

The base airfoil produced a lift force of roughly 0.153 N, while the highest lift force produced by any other airfoil was only 0.07 N, less than half of the base airfoil's. The lift and drag of each airfoil is shown in Table 1.

4 Discussion

Throughout the experiments, a general trend showed that all dimples and grooves lessened the lift force, dimples lessened the drag force, and grooves increased the drag force. Although some of this was expected, it did not completely follow the hypothesis.

The inspiration for this experiment was a golf ball, which has many dimples on it designed to increase lift and decrease drag. Because of this, it was thought that the trend would stay the same especially with the dimpled airfoils. Although the dimples lessened drag force, what was not expected was such a drastic decrease in the lift force. After looking into the mechanics of a golf ball, it was observed that another reason for the increase in lift and the decrease in drag was the centrifugal force acting on a golf ball after it was hit. The airfoils tested were at a neutral position throughout all of the testing, which means that there is no centrifugal force acting on the airfoil in the wind tunnel. This was an overlooked difference between the two use cases of surface dimples.

Another difference is that golf balls are constantly rotating as they move through the air. This varying angle of attack on the ball causes more air to push upward on the golf ball than downward. However, the airfoil had a static angle of attack. This means that the wind was not able to "pick up" the airfoil in order to increase the lift force. Another cause for the reduced lift force could be the fact that the surface was much rougher. Having dimples or grooves disrupts the smooth boundary layer, which induces turbulent flow, which is much less efficient than the laminar flow created by a smooth surface. This disruption causes the air to flow across the airfoil in a much more erratic manner, hence lessening our lift force.

The final reason for the decreased lift force was the fact that the pressure difference from the top of the airfoil to the bottom was lessened. On a normal aircraft wing or airfoil, the height difference from top to bottom is crucial in creating an overall pressure difference across the surface. This difference is reduced when there is any sort of indentation on the airfoils, causing a similar reduction in lift.

5 Conclusion

The original hypothesis that grooves and dimples would increase lift force and decrease drag force was rejected after analysis of the experiment data. As seen from the tests, the indentation on the airfoils all decreased lift, the dimples decreased drag, and the grooves increased drag. Although our hypothesis would most likely work for other spherical objects like golf balls, it did not apply for grooves and dimples on an airfoil.

It is also possible that creating a greater angle of attack than what we tested with, fifteen degrees, could lead to better results where lift would be increased. One thing that we can conclude from our experiment is that dimpling does decrease drag whether that be on golf balls or airfoils.

Further testing could include varying the angles of attack on the airfoils and the wind speed they were tested at. These other variables could have relationships to lift and drag not seen in this experiment.

4 *ME 103*

6 Acknowledgments

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7 References

- Gabrić, D. (2020, August 20). How Golf Balls Are Made. <https://branded.disruptsports.com/blogs/blog/how-golf-balls-are-made>
- Glazer, N. (2025). Understanding Bernoulli's Principle: Key Concepts Explained. <https://www.pilotmall.com/blogs/news/what-is-bernoullis-principle>
- American, S. (2005, September 19). How do dimples in golf balls affect their flight? Scientific American. <https://www.scientificamerican.com/article/how-do-dimples-in-golf-ba/>
- What Is Profile Drag? Form Drag vs. Skin Friction. (2025, October 9). Engineer Fix. <https://engineerfix.com/what-is-profile-drag-form-drag-vs-s>

8 Appendix

The next page contains photos of the airfoils used in this experiment.



Figure 6. Smallest Dimples - 1081 circular dimples



Figure 7. Medium Dimples - 253 circular dimples



Figure 8. Large Dimples - 105 circular dimples

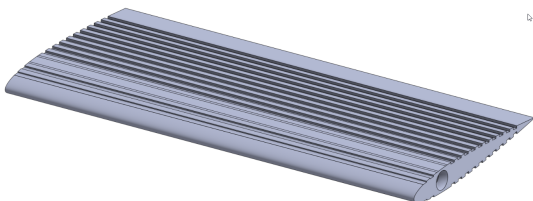


Figure 9. Small Grooves - 0.125 in grooves in airfoil

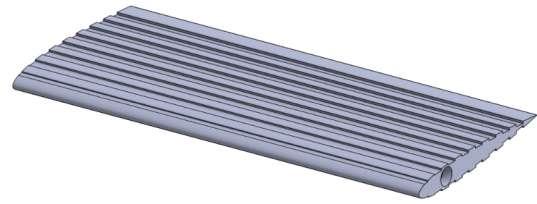


Figure 10. Medium Grooves - 0.25 in grooves in airfoil

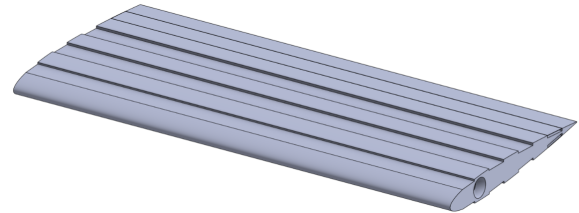


Figure 11. Large Grooves - 0.5 in grooves in airfoil

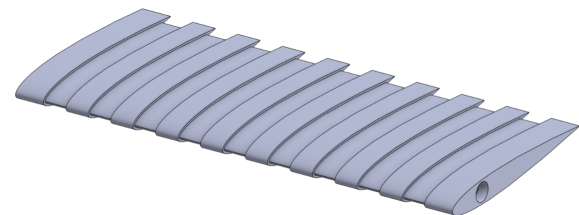


Figure 12. Sideways Grooves - 0.5 in grooves in airfoil perpendicular to length